Answer Key & Teacher Notes

Session 1 — Writing Repeated Factors as Powers · Session 2 — Finding the Value of a Power

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Writing Repeated Factors as Powers

TALK PROMPTS

- Why do we say a power shows repeated multiplication and not repeated addition?
- Your partner says the exponent counts the multiplication signs. Is that true? What does it really count?

PARTNER TASK

Write and Expand. One partner writes $7 \times 7 \times 7 \times 7$ as a power and names the base; the other checks by expanding it back. Switch roles with $4 \times 4 \times 4 \times 4 \times 4$, then compare 4^3 and 3^4 in expanded form.

WATCH FOR

- Switching the base and exponent ($4 \times 4 \times 4 \times 4 \times 4$ written as 5^4).
- Using the number of × signs as the exponent ($8 \times 8 \times 8$ written as 8^2).
- Reading 12^4 as 12×4.

SESSION 2 · TEACHER NOTES

Finding the Value of a Power

TALK PROMPTS

- What pattern do you see in the powers of 2 as the exponent grows by one? Why does it happen?
- 4^4 has the smaller base and 9^3 has the larger base. Which value is greater, and why?

PARTNER TASK

Storage Cube and Decimal Powers. One partner expands 4^3 and explains why it is not 4×3; the other evaluates it (64). Switch roles to evaluate 1.2^2 (1.44) and 0.3^3 (0.027), checking decimal places.

WATCH FOR

- Multiplying the base by the exponent ($10^3 = 30$).
- Placing the decimal point wrong in decimal powers ($0.6^2 = 3.6$ instead of 0.36).
- Thinking the larger base always gives the larger value (5^2 vs. 2^5).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.A.1	1	2	1	1	2	2	3	2	2	1	1
S1 · Form B	6.EE.A.1	1	2	1	1	2	2	3	2	2	1	1
S2 · Form A	6.EE.A.1	1	2	1	1	2	2	2	3	2	1	1
S2 · Form B	6.EE.A.1	1	2	1	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

power: A short way to write the same **factor** multiplied again and again.

base: The number that is **repeated** as a factor.

exponent: The raised number that tells how many **times** the base is a factor.

- The **base** is the repeated factor.
- The **exponent** tells how many times the base is used as a factor.
- The exponent counts factors, not **multiplication** signs.

Example: The repeated factor is **2**. That is the base. Count the factors: there are **4**. That is the exponent. Write the base with a raised exponent: 2^4. Answer: 2^4, read "2 to the **fourth** power"

YOU TRY

1. a. 6^3 b. $(1/4)^2$

2. $10 \times 10 \times 10 \times 10$; 4 factors

EXIT TICKET

1. a. 8^5 b. 0.3^2

2. base 9, exponent 6; $9 \times 9 \times 9 \times 9 \times 9 \times 9$

3. **B** (4^3)

PRACTICE FORM A

1. a. 7^4 b. 4^5 c. 5^3 d. 8^2 e. $(3/4)^2$ f. 0.4^3

2. a. 2^6 b. 10^2 c. base 2, exponent 6 d. The exponent counts how many times 2 is used as a factor.

3. 5^4; base 5, exponent 4

4. 13^3; base 13, exponent 3

5. Rafi is wrong. There are three factors of 8, so the exponent is 3: $8 \times 8 \times 8 = 8^3$.

6. a. $(1/2)^3$ b. $11 \times 11 \times 11$

7. a. 2^5 b. $2 \times 2 \times 2 \times 2 \times 2 = 32$ layers c. Each fold adds another factor of 2, so the exponent grows by 1 and the layers double each time.

8. a. $4^3 = 4 \times 4 \times 4$; $3^4 = 3 \times 3 \times 3 \times 3$ b. 3^4 (it has 4 factors; 4^3 has 3)

9. The exponent 4 means use 12 as a factor 4 times, not multiply 12 by 4. $12^4 = 12 \times 12 \times 12 \times 12$ (which equals 20,736).

10. **B** (3^5)

11. **A** (6)

Self-check word **EXPONENT**

PRACTICE FORM B

1. a. 6^3 b. 9^4 c. 11^2 d. 5^5 e. $(2/3)^3$ f. 0.7^2

2. a. 3^5 b. 12^2 c. base 3, exponent 5 d. The exponent counts how many times 3 is used as a factor.

3. 4^6; base 4, exponent 6

4. 15^4; base 15, exponent 4

5. Lena is wrong. There are four factors of 6, so the exponent is 4: $6 \times 6 \times 6 \times 6 = 6^4$.

6. a. $(1/5)^2$ b. $20 \times 20 \times 20$

7. a. 3^4 b. $3 \times 3 \times 3 \times 3 = 81$ bacteria c. Each period adds another factor of 3, so the exponent grows by 1 and the count triples each time.

8. a. $2^5 = 2 \times 2 \times 2 \times 2 \times 2$; $5^2 = 5 \times 5$ b. 2^5 (it has 5 factors; 5^2 has 2)

9. The exponent 5 means use 3 as a factor 5 times, not multiply 3 by 5. $3^5 = 3 \times 3 \times 3 \times 3 \times 3$ (which equals 243).

10. **B** (3)

11. **A** (times the base is used as a factor)

Self-check word **MULTIPLY**

GUIDED NOTES

evaluate: Find the one **number** an expression equals.

squared: To the 2nd power; use the base as a factor **2** times.

cubed: To the 3rd power; use the base as a factor **3** times.

- To evaluate a power, write the **expanded** form and multiply one factor at a time.
- $4^3 = 4 \times 4 \times 4 = 64$, not $4 \times 3 = \mathbf{12}$.
- Fraction and decimal bases work the **same** way: $(1/2)^2 = 1/4$.

Example: Expand: $6^3 = 6 \times 6 \times \mathbf{6}$. Multiply the first two: $6 \times 6 = \mathbf{36}$. Multiply by the last 6: $36 \times 6 = \mathbf{216}$. Answer: $6^3 = \mathbf{216}$, not $6 \times 3 = 18$

YOU TRY

1. a. 81 b. 0.25

2. $3 \times 3 \times 3 = 27$ cm³

EXIT TICKET

1. a. 25 b. 16

2. $6 \times 6 \times 6 = 216$ in³

3. **C** (3^3)

PRACTICE FORM A

1. a. 144 b. 27 c. 16 d. 121 e. 1.44 f. 1/9

2. $2^1 = 2 = 2$; $2^2 = 2 \times 2 = 4$; $2^3 = 2 \times 2 \times 2 = 8$; $2^4 = 2 \times 2 \times 2 \times 2 = 16$. Each value is 2 times the one before, because each new power has one more factor of 2.

3. $4 \times 4 \times 4 = 64$ cubic units (not $4 \times 3 = 12$)

4. $5 \times 5 \times 5 \times 5 = 625$

5. The poster is wrong. $7^2 = 7 \times 7 = 49$, not $7 \times 2 = 14$.

6. $2^5 = 32$, $5^2 = 25$, $3^3 = 27$; order: 5^2, 3^3, 2^5

7. a. $0.6 \times 0.6 = 0.36$ b. $1.5 \times 1.5 = 2.25$

8. a. $3^4 = 81$ cells b. $3^5 = 243$ cells c. $243 - 81 = 162$ more cells

9. Teo multiplied the base by the exponent. 10^3 means $10 \times 10 \times 10 = 1{,}000$.

10. **C** (32)

11. **A** (squared)

Self-check word **SQUARED**

PRACTICE FORM B

1. a. 81 b. 8 c. 25 d. 169 e. 1.69 f. 4/25

2. $3^1 = 3 = 3$; $3^2 = 3 \times 3 = 9$; $3^3 = 3 \times 3 \times 3 = 27$; $3^4 = 3 \times 3 \times 3 \times 3 = 81$. Each value is 3 times the one before, because each new power has one more factor of 3.

3. $5 \times 5 \times 5 = 125$ cubic units (not $5 \times 3 = 15$)

4. $6 \times 6 \times 6 = 216$

5. The poster is wrong. $8^2 = 8 \times 8 = 64$, not $8 \times 2 = 16$.

6. $3^4 = 81$, $4^3 = 64$, $2^7 = 128$; order: 4^3, 3^4, 2^7

7. a. $0.4 \times 0.4 = 0.16$ b. $2.5 \times 2.5 = 6.25$

8. a. $4^3 = 64$ tiles b. $4^4 = 256$ tiles c. $256 - 64 = 192$ more tiles

9. Malik multiplied the base by the exponent. 9^3 means $9 \times 9 \times 9 = 729$.

10. **D** (1,000)

11. **C** (36)

Self-check word **FACTORS**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · WRITING REPEATED FACTORS AS POWERS

1. a. 4^3 b. 9^2 c. 2^7 d. 10^4 e. $(1/3)^3$ f. 0.5^2 g. 6^5 h. $(2/5)^2$

2. Row 1: base 8, exponent 3, $8 \times 8 \times 8$. Row 2: 7^4, $7 \times 7 \times 7 \times 7$. Row 3: 11^2, base 11, exponent 2.

3. 3^5

4. He switched the base and exponent. The repeated factor 4 is the base, and it is used 5 times, so the power is 4^5.

SHEET 2 · FINDING THE VALUE OF A POWER

1. a. 49 b. 81 c. 125 d. 100,000 e. 64 f. 1

2. a. 1/8 b. 0.09

3. a. 3^3 b. 10^4

4. $7 \times 7 \times 7 = 343$ in³

5. A and C: $2^6 = 64$ and $4^3 = 64$ ($6^2 = 36$, $3^4 = 81$)

6. **C** (64)

7. **A** (10^3)

October 2026 conceptual additions — teacher key

Session 1 · Writing Repeated Factors as Powers

Repeated factors give $3^4 = 81$. The others both equal 12, but use multiplication and repeated addition rather than four factors of 3.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Repeated factors give $3^4 = 81$. The others both equal 12, but use multiplication and repeated addition rather than four factors of 3.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Finding the Value of a Power

$2^5 = 32$; $5^2 = 25$. No, base and exponent play different roles.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: $2^5 = 32$; $5^2 = 25$. No, base and exponent play different roles.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.